

AD 502 – Machine Learning

Pre-Reading Material – Lecture 10

Data Augmentation

Estimated Reading Time: 25–30 minutes

Why This Topic Matters

Machine learning models, especially deep learning models, often require large amounts of high-quality training data. **Data augmentation (DA)** generates additional training examples by applying meaningful transformations to existing data, helping address limited training data and improve model performance.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the concept and purpose of **data augmentation**.
- Identify common data augmentation operators.
- Understand augmentation for **text and structured data**.
- Explain interpolation and generation-based augmentation.
- Understand learning-based augmentation and automatic augmentation policies.
- Relate data augmentation to semi-supervised and active learning.

Activate Your Prior Knowledge

1. Limited Training Data

Suppose you have only 1,000 labeled examples for training a machine learning model.

Think: Can we create additional useful training examples without collecting and labeling thousands of new samples?



2. Meaning-Preserving Changes

If a sentence is modified slightly but keeps the same meaning, it may still be useful for training.

Think: Can such modified examples help a model become more robust?

3. Data Quality

Generating more data does not automatically guarantee better performance.

Think: What could happen if the augmentation changes the original meaning or introduces incorrect information?

Prerequisite Concepts

You should be familiar with:

- Machine learning training data
- Supervised learning
- Classification
- Feature representation
- Natural Language Processing basics
- Basic deep learning concepts

Core Concepts

1. What is Data Augmentation?

Data augmentation is a **data transformation technique used to generate additional training examples** from existing data. It is particularly useful when obtaining large amounts of labeled data is difficult or expensive.

2. Why is Data Augmentation Needed?

Deep learning models generally benefit from sufficient training data. However, collecting and labeling data can require significant time, cost, and domain expertise. Data augmentation provides a way to generate additional examples from existing data.

3. Basic Data Augmentation Operators

Common augmentation operations include **addition, deletion, replacement, and swapping**. The appropriate operation depends on the type of data and the task.

4. Data Augmentation for NLP

For text data, common operations include:

- **Word Replacement:** Replace a word with another suitable word.
- **Word Insertion:** Insert a new word.
- **Word Deletion:** Remove a word.
- **Word Swapping:** Exchange the positions of words.

These operations aim to create new examples while preserving the relevant meaning of the original text.

5. Token-Level and Span-Level Augmentation

Text augmentation can operate at different levels. Token-level operations modify individual words or tokens, while span-level operations can replace a larger sequence of text.



6. Interpolation-Based Augmentation

Interpolation-based approaches create new training examples by combining or interpolating existing examples. **MixUp** is a well-known example of this type of augmentation.

7. Generation-Based Augmentation

Generation-based approaches use models to generate new examples. Techniques can include **conditional generation**, **GAN-based generation**, and **sequence-to-sequence models**.

8. Learning-Based Data Augmentation

Instead of manually selecting augmentation operations, machine learning can be used to **learn which transformations or combinations of transformations are useful** for a particular task.

9. AutoAugment

AutoAugment treats the selection of augmentation operations as a policy-learning problem. A policy model learns augmentation strategies that can improve the performance of a target model.

10. Data Augmentation and Semi-Supervised Learning

Data augmentation can be used in **semi-supervised learning** as consistency training. Multiple augmented versions of an example can be generated, and the model is encouraged to produce consistent predictions for these versions.



11. Data Augmentation and Active Learning

In active learning, augmentation can help estimate uncertainty by generating multiple augmented versions of an unlabeled example and comparing the model's predictions across them.

12. Data Augmentation and Weak Supervision

Data augmentation can also work with weak supervision, where noisy sources such as **knowledge bases, crowdsourcing, or user-defined heuristics** provide supervision signals for unlabeled data.

13. Pre-trained Models for Data Augmentation

Pre-trained language models can support text augmentation through methods such as **learned token replacement and conditional generation**. Examples include approaches using models such as BERT, BART, and T5.

14. Challenges of Data Augmentation

Data augmentation introduces additional choices and hyperparameters. When many augmentation operators are combined, the search space can become large, making it difficult to identify the best augmentation strategy manually.

Guided Reading Questions

1. What is data augmentation?
2. Why is data augmentation useful when labeled data is limited?
3. What are the basic data augmentation operators?
4. What are common augmentation techniques for NLP?
5. What is the difference between token-level and span-level augmentation?
6. What is interpolation-based augmentation?



7. What is generation-based augmentation?
8. What is learning-based data augmentation?
9. How does AutoAugment work?
10. How can data augmentation support semi-supervised and active learning?

Reflection Activity

Write 100–150 words explaining how data augmentation can help when collecting and labeling new training data is expensive. Consider both the **benefits and risks** of generating additional training examples.

Self-Assessment

1. Data augmentation generates additional training examples through data _____.
2. Word replacement, insertion, and deletion are examples of augmentation for _____ data.
3. Exchanging the positions of two words is called word _____.
4. MixUp is an example of _____-based augmentation.
5. GANs can be used for _____-based data augmentation.
6. AutoAugment learns an augmentation _____.
7. Data augmentation can be used as consistency regularization in _____-supervised learning.
8. Active learning can use augmentation to estimate model _____.
9. Knowledge bases and crowdsourcing can provide supervision signals in _____ supervision.
10. A major challenge of combining many augmentation operators is the large _____ space.

Key Takeaway

Data augmentation creates additional training examples by transforming existing data. It can reduce dependence on expensive labeled datasets and improve model robustness, while advanced approaches such as **MixUp, generation-based methods,**



AutoAugment, and learning-based augmentation provide increasingly sophisticated ways to create useful training data.

